

KARAN RAJENDRA

SOFTWARE ENGINEER

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EDUCATION

Stony Brook University <i>Master of Science in Computer Engineering — GPA: 3.83/4.0</i> Coursework: Distributed Systems, Operating Systems, Computer Networks, Advanced Algorithms	Aug 2023 – May 2025 <i>Stony Brook, NY</i>
Visvesvaraya Technological University <i>Bachelor of Engineering in Computer Science</i>	Aug 2017 – Aug 2022 <i>Bangalore, India</i>

WORK EXPERIENCE

Research Assistant Stony Brook University - Developed automated test and preprocessing pipelines for high-resolution medical datasets, reducing runtime by 25% with optimized data structures and parallelization. - Optimized ML training workflows using modular design and mixed-precision computing, conducting performance testing and profiling (perf, Valgrind) to cut experiment time by 20%. - Deployed production ML pipelines across local, HPC, and cloud environments with Git, Docker, and CI/CD, supporting clinical diagnostic workflows. - Leveraged PyTorch & TensorFlow for experimentation, integrating reproducibility and performance monitoring.	Jun 2024 – May 2025 <i>Stony Brook, NY</i>
Graduate Teaching Assistant – Computer Vision Stony Brook University - Mentored 40+ graduate students in Computer Vision and Deep Learning, supporting projects with Python, OpenCV, and industry-grade coding practices. - Designed and graded assignments on modular design, debugging, and algorithm optimization, boosting project quality by 25% and cutting runtime errors by 15%.	Aug 2024 – Dec 2024 <i>Stony Brook, NY</i>
Software Engineer Impavid Technologies - Redesigned legacy fare computation into modular REST APIs, reducing response time to 120ms and improving throughput during peak hours. - Optimized a Cassandra-backed analytics store for real-time surge pricing, enabling dynamic fare adjustments across 12+ metropolitan zones through profiling and performance validation. - Resolved 15+ production incidents with CloudWatch Logs & ELK Stack, applying system-level debugging and root-cause analysis to improve reliability and reduce downtime. - Built a React.js onboarding dashboard with 18+ reusable components, cutting onboarding time and improving design fidelity through close UX/QA collaboration. - Deployed containerized microservices on AWS EKS, automating deployments with Docker & Kubernetes to support scalable system rollouts.	Sep 2022 – Jun 2023 <i>Bangalore, India</i>

PROJECTS

Graph Partitioning – FM Algorithm - Implemented the Fiduccia–Mattheyses (FM) algorithm in C/C++ with a bucket-based gain structure, reducing edge cut size by 62.7% on industry-standard IBM test circuits. - Optimized runtime to O(n) by designing efficient data structures and applying profiling/debugging (GDB, Valgrind) to enhance scalability for large-scale circuit graphs.	C/C++, Algorithms, Data Structures
VLSI Floorplanning – Timberwolf Algorithm - Designed the Timberwolf placement algorithm with Simulated Annealing in C/C++, reducing average wirelength by 18% and minimizing overlap on IBM benchmark datasets. - Constructed robust data structures (Node, Net, Grid, Result) and cost functions, improving placement quality and routability over baseline heuristics.	C/C++, Algorithms, Optimization, Distributed Systems

SKILLS

Languages:	C, C++, Python, JavaScript (ES6+), TypeScript
CS Fundamentals:	Data Structures & Algorithms, OOP
Web/Backend:	React.js, Next.js, Redux, Node.js, Express.js, REST APIs
Databases/Cloud:	MySQL, PostgreSQL, MongoDB, AWS (EC2, S3, Lambda), Docker, GitHub Actions, Kafka
Other/ML:	Git/GitHub, Linux, Bash, Debugging (GDB, Valgrind, perf), Unit Testing (Jest, PyTest), CI/CD, JWT/OAuth